

Chit 15 — MongoDB: Aggregation and Indexing (Department/Teacher)

1 & 2. Create database and teacher collection

```
use department
db.teacher.insertMany([
  { name:'Arun', department:'CS', experience:5, salary:20000 },
  { name:'Meena', department:'CS', experience:8, salary:30000 },
  { name:'Ravi', department:'IT', experience:3, salary:18000 },
  { name:'Sita', department:'IT', experience:10, salary:25000 },
  { name:'Kiran', department:'Maths', experience:6, salary:15000 }
])
```

3. Department-wise average salary

```
db.teacher.aggregate([
  { $group: { _id: '$department',
              avg_salary: { $avg: '$salary' } } }
])
```

4. Number of employees in each department

```
db.teacher.aggregate([
  { $group: { _id: '$department',
              count: { $sum: 1 } } }
])
```

5. Department-wise minimum salary

```
db.teacher.aggregate([
  { $group: { _id: '$department',
              min_salary: { $min: '$salary' } } }
])
```

6. Apply index and drop index

```
// Create index
db.teacher.createIndex({ department: 1 })

// Show all indexes
db.teacher.getIndexes()

// Drop specific index
db.teacher.dropIndex({ department: 1 })

// Confirm drop
db.teacher.getIndexes()
```